

Lab: What Does Surprise Feel Like?

Objective

Investigate perception as an embodied, active process by exploring how the body generates predictions, encounters surprise, and updates its model of the world. This lab makes prediction error tangible through direct sensory experience.

Prerequisites

- A quiet space for 25 minutes
- A few small objects of different textures and temperatures
- Optional: a blindfold or sleep mask
- No prior knowledge required

Part 1: The Body Already Knows (5 minutes)

Sit with your eyes closed. Bring your attention to your feet on the floor. Without moving, notice: your body already has a detailed prediction of what the floor feels like -- its hardness, temperature, texture. You are not perceiving the floor from scratch each moment; your body is predicting it and only updating when something changes.

Now wiggle your toes. Notice how the sensation shifts. The body's prediction was disturbed by movement, generating fresh sensory information. In Active Inference, this is the sensorimotor contingency: perception changes with action.

Write your response here

(Describe what you noticed about the floor before and after wiggling your toes. What was predicted versus what was new?)

Part 2: Surprise Mapping (7 minutes)

With eyes closed or blindfolded, have someone place three objects in front of you (or arrange them yourself without looking). Reach for each one slowly.

For each object: 1. Before touching, notice: What does your body predict it will feel? 2. At the moment of contact: Does the sensation match? Where in your body do you register the match or mismatch? 3. After a few seconds of holding: How quickly does the surprise fade as the body updates its model?

If you are alone, reach for objects in your environment without looking -- the edge of a table, a fabric surface, a glass.

Write your response here

(For each object, describe: (a) what you predicted, (b) what you actually felt, and (c) where in your body you registered the surprise or confirmation.)

Part 3: Sensorimotor Exploration (5 minutes)

This exercise demonstrates that perception is constituted by movement (Noë, O'Regan).

1. Hold a textured object (a key, a coin, a piece of fabric). Keep your fingers still against it. Notice how the texture perception begins to fade.
2. Now move your fingers across the surface. The texture comes alive again. Movement generates perception.
3. Try this with your eyes: Stare at a single point without moving your eyes. After 20-30 seconds, notice how the periphery begins to fade. Now let your eyes move naturally. The world returns.

Perception is not passive reception -- it is enacted through the body's movement.

Write your response here

(Describe what happened when you held still versus when you moved. How does movement create perception?)

Part 4: Interoceptive Surprise (5 minutes)

Surprise is not limited to external senses. The body also generates predictions about its internal states.

1. Take three rapid, deep breaths. Notice the bodily surprise -- dizziness, tingling, altered heart rate. Your body predicted a normal breathing pattern and encountered something unexpected.
2. Hold your breath gently for 10-15 seconds (do not strain). Notice the growing urge to breathe. This is interoceptive prediction error: the body predicted continued oxygen supply and is now registering a mismatch.
3. Resume normal breathing. Notice how quickly the body returns to its predicted baseline.

Write your response here

(Describe the bodily sensations during each perturbation. Where did you feel the interoceptive surprise? How did the body recover?)

Part 5: Reflection

Write your response here

(How is perception different from what you thought before this lab? What does it mean that the body is always predicting, and perception is what happens when those predictions meet reality?)

Reflection Table

Perceptual Mode	What I Predicted	What I Actually Felt	Surprise Level
Touch (object)			
Movement and texture			
Vision (staring)			
Breath perturbation			